Support for Cirq MatrixGates of >2 qubits and ControlledGate

https://github.com/quantumlib/qsim/issues/163

ROW 74

Support for Cirq MatrixGates of >2 qubits and ControlledGate #163

New issue

Closed

andrewyatesg opened on Aug 1, 2020

I noticed that the qsim Cirq interface doesn't support MatrixGates that act on more than two qubits - can we get support for this?

I also noticed that if I create a controlled gate via something like this:

`ControlledGate(MyGate)(ctrl_qubit, *target_qubits)`

then qsim fails. For example `ControlledGate(cirq.X)(qubits[0], qubits[1])` returns the error

`ValueError: Operation doesn't satisfy the given 'keep' but can't be decomposed: cirq.X(cirq.LineQubit(1)).controlled_by(cirq.LineQubit(0))`.

(I know that in this situation, one should use cirq.CX.) Can we get support for arbitrary ControlledGates? And are there any workarounds for the above two issues? Thanks!

95-martin-orion on Aug 3, 2020 · Collaborator

Unfortunately, the two-qubit matrix gate limitation is based on a core assumption of qsim that *all* gates act on at most two qubits. In my understanding, extending this to 3-qubit gates might be possible, but it would involve compromises in performance on 1- and 2-qubit gates.

Aside from manually decomposing these gates into 1- and 2-qubit gates, I don't know of a good workaround for this - **@sergeisakov** might be able to explain further.

On a similar note, controlled gates on >2 qubits (including control qubits) are also out of scope for qsim. Controlled 1-qubit gates should be possible, but currently Cirq doesn't support removing `controlled_by` via gate decomposition. In theory we could handle this in qsim, but I think the correct solution would instead be to update the decomposition behavior in Cirq. This is an area of active discussion in the Cirq dev team - I encourage you to open an issue there.

Until that's resolved, the workaround for controlled 1-qubit gates would be to use builtin controlled gates where possible, and define any others as 2-qubit matrix gates. Larger controlled gates would have to fall back to manual decomposition.

we-taper on Aug 6, 2020

Hi **@95-martin-orion** , I wonder if it is possible to add subroutines specifically for multi-qubit controlled gates and also prevent compromises in performance on 1 and 2 qubit gates? There is a very simple algorithm that can simulate gates of very large controlled qubits with very small memory overhead. Is it possible to integrate this to `qsim` ?

sergeisakov on Aug 6, 2020 · Collaborator

There are plans to add support for multi-qubit gates in general and for multi-qubit controlled gates in particular. It will be added at some point later.

MichaelBroughton mentioned this on Aug 21, 2020

Assignees
No one assigned

Labels
No labels

Type
No type

Projects
No projects

Milestone
No milestone

Relationships
None yet

Development
No branches or pull requests

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Participants

+1

MichaelBroughton mentioned this on Aug 21, 2020

Allow custom gate serialization tensorflow/quantum#354

95-martin-orion mentioned this on Aug 26, 2020

Attempt to decompose Cirq gates #183

we-taper on Aug 31, 2020

Hi qsim developers, we are filling this RFC for an implementation of multi-controlled 1 qubit gates.

Looking forwards to comments!

sergeisakov on Aug 31, 2020 Collaborator

Thank you for writing that. We are actually working on multi-qubit and multi-controlled gates. It is likely that multi-controlled gates will be implemented in a different way from as implemented in the QuEST simulator.

MichaelBroughton on Oct 22, 2020 Collaborator

@sergeisakov any updates on progress here ? This would definitely be a nice to have feature for TFQ.

95-martin-orion on Oct 22, 2020 Collaborator

PR #216 allows the C++ layers of qsim to recognize (but not execute) multi-controlled gates. The effort to support execution of these gates, as well as translating them from Cirq, is ongoing.

yiyou-chen on Nov 3, 2020 · edited by yiyou-chen Edits

Got this error when doing controlled-ry:
ValueError: Operation doesn't satisfy the given `keep` but can't be decomposed:
cirq.ry(np.pi*0.8904292526573554).on(cirq.GridQubit(0, 1)).controlled_by(cirq.GridQubit(0, 0))
would be really helpful if it can do 2-qubit controlled gates like C-ry(theta) if not multi-controlled or multi-qubit gates.
Thanks.

95-martin-orion on Nov 3, 2020 Collaborator

Thanks for highlighting this, @4Dmovie. Work on this issue has been focused on multi-controlled gates in order to provide a general solution to the problem, with Sergei's PR #231 being the most recent change submitted. I'll follow up with the qsimcirq support shortly, and will update this issue when the feature is available.

👍 1

95-martin-orion mentioned this on Nov 3, 2020

Allow control passthrough from Cirq. #233

95-martin-orion mentioned this on Nov 3, 2020

Allow control passthrough from Cirq. #233

95-martin-orion on Nov 3, 2020 — Collaborator

With PR #233, controlled gates requested in Cirq should now be able to take advantage of the support in qsim-core. In particular, @4Dmovie's example with controlled-ry should work at HEAD (but note that we haven't released this change, so you'll need to install directly from GitHub to pick it up).

Support for larger matrix gates is still in progress.

❤️ 1

yiyou-chen on Nov 4, 2020

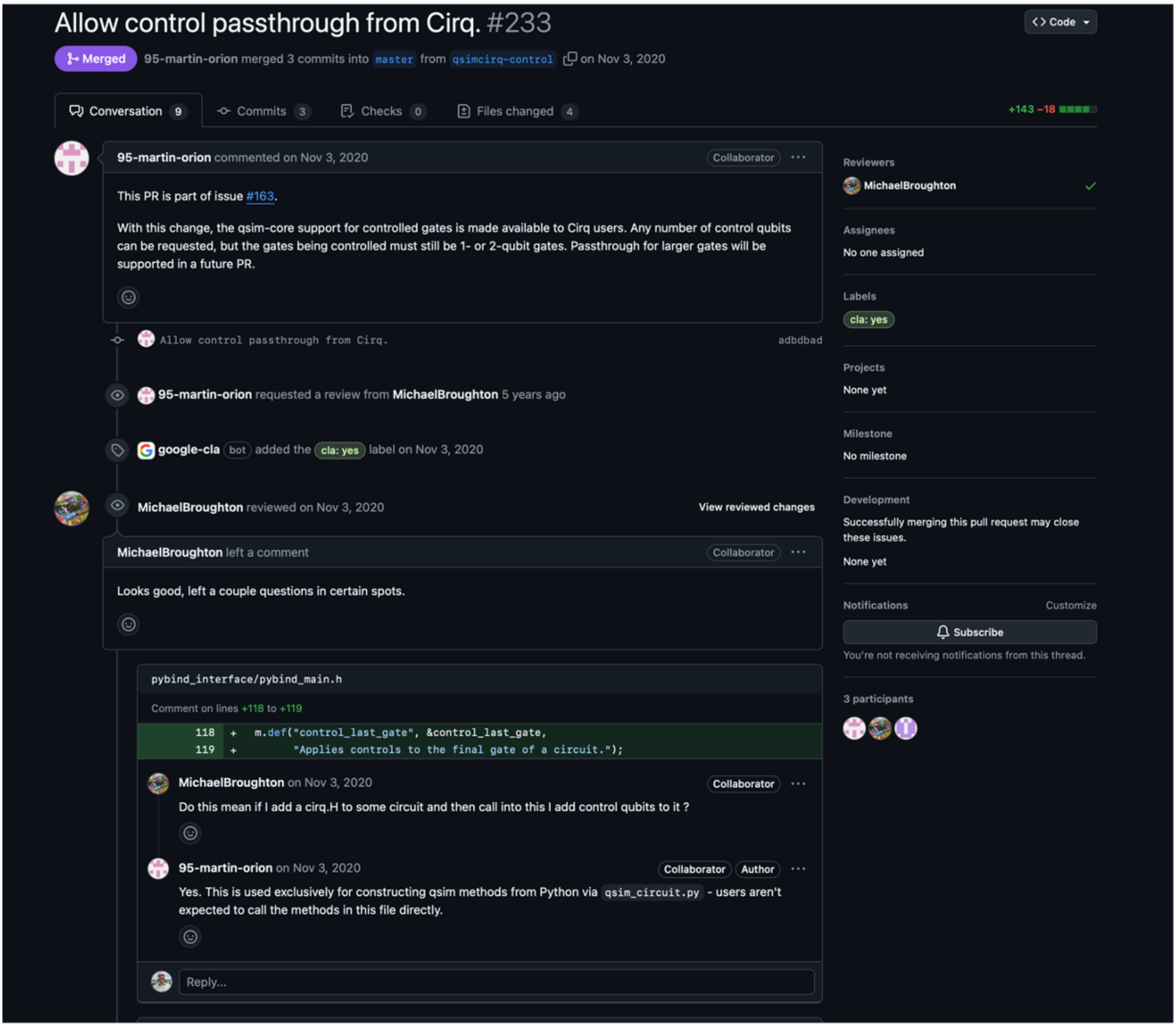
> With PR #233, controlled gates requested in Cirq should now be able to take advantage of the support in qsim-core. In particular, @4Dmovie's example with controlled-ry should work at HEAD (but note that we haven't released this change, so you'll need to install directly from GitHub to pick it up).
>
> Support for larger matrix gates is still in progress.

Works for C-Ry now. Thank you so much for your team's amazing work! I'll let you know if further functionalities are demanded.

95-martin-orion on Dec 3, 2020 — Collaborator

qsimcirq v0.6.0 provides support for arbitrary gates of up to six qubits. This includes any gate that can decompose into the qsim gateset (defined in gates_cirq) as well as any gate that has a matrix definition.

95-martin-orion closed this as completed on Dec 3, 2020


Allow control passthrough from Cirq. #233

Merged 95-martin-orion merged 3 commits into `master` from `qsimcirq-control` on Nov 3, 2020

Conversation 9 | Commits 3 | Checks 0 | Files changed 4 | +143 −18

95-martin-orion commented on Nov 3, 2020 — Collaborator

This PR is part of issue #163.

With this change, the qsim-core support for controlled gates is made available to Cirq users. Any number of control qubits can be requested, but the gates being controlled must still be 1- or 2-qubit gates. Passthrough for larger gates will be supported in a future PR.

`Allow control passthrough from Cirq.` adbdbad

95-martin-orion requested a review from **MichaelBroughton** 5 years ago

google-cla bot added the cla: yes label on Nov 3, 2020

MichaelBroughton reviewed on Nov 3, 2020 — View reviewed changes

MichaelBroughton left a comment — Collaborator

Looks good, left a couple questions in certain spots.

`pybind_interface/pybind_main.h`

Comment on lines +118 to +119

```
118 +   m.def("control_last_gate", &control_last_gate,
119 +         "Applies controls to the final gate of a circuit.");
```

MichaelBroughton on Nov 3, 2020 — Collaborator

Do this mean if I add a cirq.H to some circuit and then call into this I add control qubits to it ?

95-martin-orion on Nov 3, 2020 — Collaborator, Author

Yes. This is used exclusively for constructing qsim methods from Python via `qsim_circuit.py` - users aren't expected to call the methods in this file directly.

Reply...

Reviewers: MichaelBroughton

Assignees: No one assigned

Labels: cla: yes

Projects: None yet

Milestone: No milestone

Development: Successfully merging this pull request may close these issues. None yet

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